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Impact Report 2021–2026

Looking back

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A message from the Director

Five years ago, whether a shopper walked out with an unscanned item, whether a stuck fare gate opened for a commuter, or which caller a queue reached next sat mostly with whoever happened to be standing nearest the decision at the time. It is worth pausing, at the start of a report like this one, on how quickly that stopped being true — and on how much of the change this School was asked to measure rather than to make.

We came to this work convinced that a system quietly making calls on someone's behalf deserved the same scrutiny as a system making a case for funding, and five years later I am less certain than I was that scrutiny alone closes the gap it opens. We walked back from at least one instrument we were confident about early on, and the walking-back belongs in this report as much as anything we got right, because a report that only counts its successes has stopped measuring anything.

What held, across five years and a great many partner floors, was the community around the counting: the staff who logged every flagged incident long after the novelty wore off, the researchers who sat with a call-centre roster until the pattern gave itself up, the collaborators who let us watch a handover happen in real time rather than reconstruct it afterwards. Each cycle of that work compounded a little on the last, the flywheel turning faster than most of us noticed until we sat down to write this.

The honest answer, on most of what follows, is still partial. It is, I think, a more useful answer than a confident one would have been.

Director, School of Continuous Improvement

Five years in numbers

The four figures the School is asked for most often, five years in.

83 collaborators drawn into the counting

From a metropolitan transit authority to a Nordic evaluation agency, partner organisations opened their own automated systems to the School's researchers over the past five years.

341 publications on trust and automation

Peer-reviewed output across the School's evaluation-ecosystem agenda, more than half of it co-authored with a partner organisation's own staff.

4,216 completions of Deciding Under Measurement

The School's Capability Sprint on automated judgement calls filled faster than any other in the line, cohort after cohort, across five years.

22 partner institutions across three continents

International collaborators contributed comparative data no single campus could have generated alone.

Who we are

The School of Continuous Improvement studies the measurement systems that improve the things they measure — and, increasingly, the systems that have quietly taken over deciding them. Our researchers work across the Adaptive Metrics Lab and a wider evaluation-ecosystem agenda that treats an indicator, a dashboard, or a scoring rule as an object worth studying in its own right, not just a tool for studying something else.

The School's flagship postgraduate programme, the Master of Applied Measurement, trains a cohort each year to build, calibrate, and eventually retire the instruments an organisation comes to depend on. Our short-course line, the Capability Sprints, carries the same instinct into workplaces that have no time for a full degree but every reason to understand what their own dashboard is doing to them.

Our people describe themselves, more often than not, as translational improvement researchers: people equally at home in a spreadsheet and on a shop floor. PhD candidates join labs organised around the counted, the countable, and the not-yet-counted, and stay — our attrition figures are unremarkable, which in this field we have learned to treat as a finding in itself.

Our work in five domains

The five areas below carry the figures behind them; where the two disagree, both are printed.

Building the instrument

Five years ago, whether a stuck fare gate opened for a commuter without a valid tap was a judgement call made by whoever was closest to the gate. The Adaptive Metrics Lab was asked to help a metropolitan transit authority replace that judgement with a graded override score — not to remove the person at the gate, but to give the override a consistent basis in the first place. What followed was three successive instrument revisions, a validation study spanning two fare-gate networks, and a slower conversation than anyone expected about which errors an override score should be allowed to make. Building an instrument that scores a stranger's plausible excuse turned out to be less an engineering problem than a values one wearing an engineering problem's clothes.

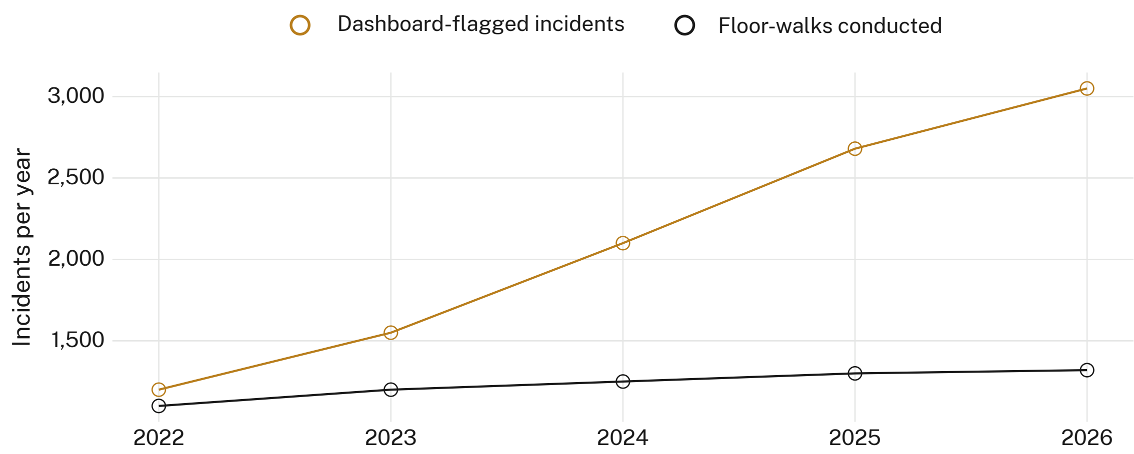
Working with the transit authority's gate-operations team, Associate Professor Beng's group spent eighteen months instrumenting a single interchange before the override score went anywhere near a live gate. Operators kept a paper log of every override they made and why, a practice the authority had abandoned years earlier and the Lab asked them to revive for the study's duration. The resulting instrument scores six factors — ticket history, gate-hesitation pattern, time of day, and three more the authority asked the School not to publish in detail — and routes only the low-confidence cases to a person. Confidence itself, the team found, needed almost as much calibration as the override decision it was meant to simplify.

> "An override score that can't say why it trusts itself isn't a score you can hand back to a person to check. Getting the confidence figure right took longer than the score underneath it."
>
> — Associate Professor Casimir Beng, Lead of the Adaptive Metrics Lab

Reading the floor from the dashboard

A self-checkout dashboard is built to save someone a walk across the shop floor, and for the most part it does. A five-year partnership with a big-four professional services firm's retail-advisory practice found a second effect nobody had asked the dashboard to produce: staff increasingly treated an unflagged terminal as a resolved terminal, whether or not anyone had actually looked at it. Dr Hanke's group, working from the Living Dashboard's methodology, tracked the growing distance between what the dashboard reported and what a physical floor-walk still found, across fortyone stores and five years of shift logs. The dashboard did not get worse at its job over that time. The floor-walk simply stopped keeping pace with it.

At one flagship store, the retail-advisory partner's own staff volunteered to run a parallel floor-walk against the dashboard for six months, checking every unflagged terminal at least once per shift regardless of what the dashboard reported. The parallel walk caught eleven incidents the dashboard had cleared as resolved, all within the first ten weeks; after that the yield fell close to zero, and the partner quietly wound the parallel walk back to its original schedule. Dr Hanke's team read that decline two ways in the report that followed: either the dashboard had genuinely closed the gap, or the floor-walk itself had drifted back into trusting what the dashboard told it. The partnership's data could not settle which.



Dashboard-flagged incidents against staff floor-walks conducted, five partner stores, 2022–2026. By 2026 the dashboard flagged 3,050 incidents against 1,320 physical floor-walks — 2.3 flags for every walk, more than double the 2022 ratio.

> "We went into the parallel walk expecting to prove the dashboard right. Six months later we weren't sure what we'd proven, which the School tells us is itself a result."
>
> — Partner, a big-four professional services firm

Who the queue trusts

A call-centre queue-routing system decides, several times a second, which caller reaches a person next and which stays in the automated branch a little longer. Dr Adeyemi's group, extending her earlier work on incident-report escalation, spent three years inside a regional university network's shared student-services call centre studying what happened when a supervisor overrode the router's decision — and, more often, what happened when a caller simply hung up before either the router or the supervisor acted. The study reframed "abandonment", the industry's own term for a caller who leaves the queue, as a decision the caller made about trust rather than a failure the router alone was responsible for.

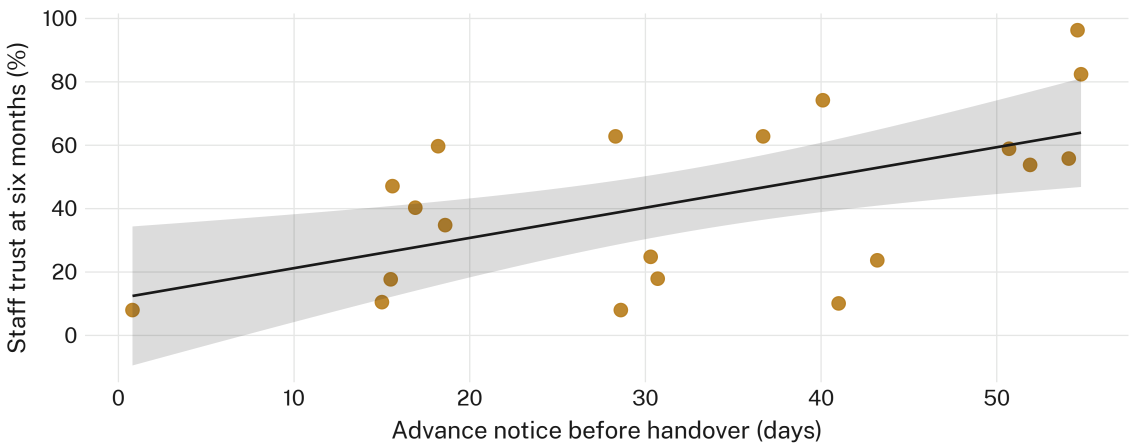
Working across four member campuses of the university network, the team logged 6,200 queue events over eighteen months, coding each one for whether a caller stayed, was routed to a person, or left before either happened. Supervisors reviewing the router's decisions in real time overrode it in roughly one call in twenty — almost always to bump a caller who had called back a second or third time that day, a signal the router did not track at all. Adding call-recurrence to the router's own inputs cut same-day repeat calls by a modest margin the network's staff called worthwhile and Dr Adeyemi's report called suggestive rather than conclusive.

> "The router already knew everything about the call in front of it and nothing about the two calls before it. Once we gave it that memory back, the calls it lost changed shape."
>
> — Dr Solveig Adeyemi, Lecturer, School of Continuous Improvement

Catching the handover early

Most organisations do not announce the day a decision moves from a person to a system; it simply starts happening more often until nobody remembers the last time a person made the call unassisted. Working with a Nordic evaluation agency's member network, Dr Vandermeer's group tested whether giving a workplace explicit advance notice of an approaching automated handover — weeks rather than a single memo — changed how much staff trusted the system six months after it arrived. The Review Cadence Observatory's usual brief is watching when a review is overdue; this extended the same instinct to watching for a handover before it happened rather than accounting for it afterwards.

Across twenty workplace pilots in the agency's network, sites that received several weeks' documented notice before an automated system took over a decision reported, on average, higher staff trust in that system at the six-month mark than sites that received a single announcement or none at all — though the relationship was loose enough that several long-notice sites landed no higher than the shortest-notice ones. Dr Vandermeer's team resisted the temptation to read the loose relationship as noise: three of the weakest-performing long-notice sites, on closer interview, had given notice of the wrong thing, flagging the tool's arrival without ever mentioning what decision it would actually make.



Advance notice given before an automated handover, in days, against staff trust in the system at six months, twenty workplace pilots (r = 0.58). Notice alone did not guarantee trust; three of the longest-notice sites clustered among the lowest scores.

> "Warning someone a system is coming is not the same as telling them what it will decide. We had to relearn that distinction three times before the notice actually worked."
>
> — Director of Evaluation, a Nordic evaluation agency

Retiring a trust score

An indicator that once told an organisation something useful about whether people trusted an automated decision does not stop being consulted just because it has stopped being accurate. Dr Sabel's contribution to the Indicator Commons over the past five years has been a working protocol for retiring exactly this kind of score — tested, at the invitation of a cooperative research consortium, against four trust indicators the consortium's member organisations had each kept running for longer than anyone could justify. The protocol does not simply switch an indicator off; it requires whoever inherits the gap to say, in writing, what they will watch instead.

One consortium member had kept a customer-trust score alive for six years after the system it originally measured had been replaced twice over, because the score fed a dashboard three other departments still checked every Monday. Dr Sabel's team's retirement protocol forced an uncomfortable meeting: the three departments, asked directly what decision the Monday number actually informed, could not agree on an answer between them. The score was retired within the month. What replaced it, six weeks later, was a shorter, less satisfying number that at least everyone in the room could explain.

> "An indicator survives contact with a dashboard for as long as someone finds it easier to keep checking than to ask what it is for. The hard part was never building the retirement protocol. It was getting anyone to use it."
>
> — Dr Renke Sabel, Convenor, Indicator Commons

Where we're headed

The next five years will ask a harder question of the School than the last five did. Where this report has mostly measured systems that already made a decision and left a trace behind them, the School's coming work will lean further into anticipation: catching a handover before it completes rather than auditing it once it has.

The Review Cadence Observatory will extend its watching brief to a wider set of partner networks, and the Living Dashboard will carry an explicit retirement clock on every indicator it hosts, so that no score outlives the question it was built to answer without someone having to notice on its behalf. The Master of Applied Measurement will introduce a dedicated stream on instrumenting automated decisions, drawing directly on the fare-gate, dashboard, and call-routing work in this report, and Capability Sprint enrolments in Deciding Under Measurement will be met, cohort permitting, with a second offering each year rather than one.

None of this will resolve the tension this report keeps returning to: that measuring a system closely enough to trust it usually means building a second system to do the measuring, and that second system will eventually need watching too. The School intends to keep building the instruments anyway, and to keep publishing the ones that told us we were wrong. A community that only measures its successes will, eventually, have nothing left worth measuring.

Our partners

The work in this report belongs to a great many people this report cannot name individually: the shop-floor staff who logged incidents long after the study that asked for them had ended, the call-centre supervisors who let a researcher sit beside them for a shift, the transit-authority operators who kept a paper log nobody else wanted to keep. We thank the partner organisations across government, industry, and the university sector who opened their automated systems to scrutiny they had no obligation to invite, and the international collaborators who brought comparative evidence no single campus could generate alone.

We thank, too, the School's professional staff, whose work rarely appears in a report shaped around research findings but without whom none of the findings would exist. And we thank the cohorts of Master of Applied Measurement students and PhD candidates who spent five years asking, in seminar rooms and site visits alike, exactly the uncomfortable questions this report has tried to answer honestly.

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